

The strong-coupling threshold made explicit: Yarotsky's proof gives $g_0^2 \sim 10^{100}$, and only a direct expansion can give $g_0^2 \sim 10^2$

The T2 theorem of this programme (T2 note) has an existential threshold $\beta \leq \beta_*(3, \{0, 1\}^3)$, with $\beta = 54/g^4$ for $SU(3)$. Following the proof of Yarotsky's Theorem 1 with numbers, the threshold it supplies is

$$\beta_* \simeq e^{-465}, \quad g_0^2 = \sqrt{54/\beta_*} \simeq 10^{101},$$

because the time-discretized cluster expansion pays a factor $e^{t_0|\Lambda_0|^2} = e^{64t_0}$ per excited site (Lemma 3 of the paper), and the classical decay e^{-t_0} per unexcited site-time must beat a polymer entropy of order 220 per site, forcing $t_0 \geq 7$. The strong-coupling region that the theorem covers, made explicit, is therefore empty for every practical purpose, and **the strong-coupling boundary of the intermediate region must be obtained from a direct expansion**, of the Kirkwood–Thomas or Datta–Kennedy type for the Hamiltonian or of the Osterwalder–Seiler character type for the Euclidean measure, whose explicit radii are of order $\beta \lesssim 10^{-2}$, that is $g_0^2 \sim 10^2$ for $SU(3)$. The width of the intermediate region in one-loop doublings is then set by the constants on both sides: about 10^3 with the crude constants now in hand ($1/g^2 \gtrsim 10^2$ on the weak side, $g^2 \gtrsim 10^2$ on the strong side), and about 20 if both sides are pushed to $g^2 \sim 1/2$ and $g^2 \sim 70$. Constants explicit; nothing promoted.

1. The chain of conditions in the proof

Yarotsky's expansion (Commun. Math. Phys. 261 (2006) 799, Section 2; passage level, from the copy on file) writes $e^{-t_0 H} = \sum_I T_{\Lambda, I}$, isolates the regions I where the perturbation acts and the regions J where classical excitations propagate, and bounds a configuration $C = \{(I_k, J_k)\}_{k=1}^N$ by

$$|w(C)| \leq \prod_{k=1}^N (2\alpha e^{t_0\beta/\alpha})^{|I_k|} e^{-t_0(|J_k| - |\Lambda_0|^2|I_k|)}$$

(Lemma 1 for the first factor, via a many-variable Schwarz lemma; Lemma 3 for the second, where the paper's displayed $|\Lambda_0|^3$ is weakened from the $|\Lambda_0|^2$ its proof establishes). The expansion is then a polymer gas on the space-time lattice $\{1, \dots, N\} \times \Lambda$, and Theorem 1 follows from “choose t_0 large and then α, β small so that $w(\chi) \leq \epsilon^{|\text{supp } \chi|}$ ” together with a bound c^n on the number of polymers of support n through a point.

For a **bounded** perturbation, $|\phi_x(v, v)| \leq \beta_0 \|v\|^2$, condition (2) holds with every $\alpha > 0$, so α is free and the activity per site of I is

$$\min_{\alpha > 0} 2\alpha e^{t_0\beta_0/\alpha} = 2e t_0\beta_0 \quad (\alpha = t_0\beta_0),$$

so the choice $\alpha = 0$ made in the T2 note is legitimate for the hypothesis of the theorem and is replaced by $\alpha = t_0\beta_0$ inside the proof.

2. The numbers

Entropy. Two sites of $\mathbb{Z}^3$ have overlapping ranges $\Lambda_0 + x, \Lambda_0 + y$ exactly when $y - x \in \{-1, 0, 1\}^3$, so the connectivity graph of the polymer gas has degree $d \leq 3 \cdot 27 - 1 = 80$ (space and the two time neighbours), and the number of connected supports of size n through a point is at most $(ed)^n$, giving $c \leq e \cdot 80 \simeq 220$.

Convergence. The Kotecký–Preiss criterion (Commun. Math. Phys. 103 (1986) 491; meta-data level) with $a(\chi) = |\text{supp } \chi|$ is implied by $\sum_{n \geq 1} (ce\rho)^n \leq 1$, where ρ is the largest per-site activity, so it suffices that

$$\rho \leq \frac{1}{2ec} \simeq 8 \times 10^{-4}.$$

The classical sites. $\rho_J = e^{-t_0} \leq 8 \times 10^{-4}$ forces $t_0 \geq 7.1$.

The perturbed sites. Each site of I carries $2e t_0 \beta_0 \cdot e^{|\Lambda_0|^2 t_0} = 2e t_0 \beta_0 e^{64 t_0}$, and with $t_0 = 7.1$,

$$\beta_0 \leq \frac{8 \times 10^{-4}}{2e \cdot 7.1} e^{-454} \simeq e^{-465}.$$

The threshold. $\beta_* \simeq e^{-465}$, so for $SU(3)$

$$g_0^4 = \frac{54}{\beta_*} \simeq 54 e^{465}, \quad g_0^2 \simeq 7.3 \times e^{232} \simeq 10^{101}.$$

Refinements inside the same proof, such as replacing the bound $|J \cap (\Lambda \setminus \tilde{I})| \geq |J| - |\Lambda_0|^2 |I|$ by an exact count of the sites shadowed by I , reduce 64 to something of order 10 and g_0^2 to something of order e^{40} . The structure of the argument, a time-discretized expansion in which every perturbed site suppresses the classical decay over its whole range for a time t_0 that must itself be large, cannot bring the threshold near $g^2 \sim 10^2$.

3. What a direct expansion gives

Three families of explicit strong-coupling results exist, and each has a radius of the form $\beta \lesssim 1/(C \cdot \text{coordination})$ with C of order one:

- *Kirkwood–Thomas* (Commun. Math. Phys. 88 (1983) 569; metadata level): a Hamiltonian expansion for the ground state of a classical gapped operator plus a bounded perturbation, written directly in the operator, with no time discretization. For the Kogut–Susskind Hamiltonian the classical part is the electric term with local gap $\varepsilon_0 = (g^2/2)C_2 \hbar c/a$ and the perturbation is the plaquette sum, each plaquette touching four links and each link lying in twelve plaquettes.
- *Datta–Kennedy* (J. Stat. Phys. 108 (2002) 373; metadata level): the same structure with explicit control of the one-quasiparticle band, which is the object the gap lower bound needs.
- *Osterwalder–Seiler* (Ann. Phys. 110 (1978) 440; metadata level): the Euclidean character expansion of the Wilson measure, convergent for small $\beta_W = 2N/g^2$, with the mass gap and clustering as consequences.

In each case the sufficient condition has the form

$$\frac{\|\text{plaquette term}\|}{\text{local gap}} \times (\text{coordination}) \times C \leq 1,$$

and with $\|\phi\|/\varepsilon_0 = 24N/(g^4 C_2) = 54/g^4$, coordination 12 and $C \simeq e$ this reads $g^4 \gtrsim 54 \cdot 12 \cdot e \simeq 1.8 \times 10^3$, that is

$$g_0^2 \sim 40\text{--}70 \quad (SU(3)),$$

in line with the rough figure used earlier. A written derivation of the Kirkwood–Thomas expansion for the Kogut–Susskind Hamiltonian with these constants would replace the existential β_* of T2 by a number of this size; that derivation is the natural next step on the strong-coupling side.

4. The width of the intermediate region

With $2b_0 \log 2 = 0.0966$ for $SU(3)$, the one-loop count of doublings between the two boundaries is $n = (1/g_{\text{weak}}^2 - 1/g_{\text{strong}}^2)/0.0966$:

weak boundary	strong boundary	n
$1/g^2 = 10^2$ (crude window)	$g^2 = 10^{101}$ (Yarotsky explicit)	$\simeq 10^3$
$1/g^2 = 10^2$	$g^2 = 70$ (direct expansion)	$\simeq 10^3$
$g^2 = 1/2$	$g^2 = 70$	$\simeq 21$

The strong-side improvement from 10^{101} to 70 changes n by less than one doubling, since $1/g^2$ is tiny at both; the width is set almost entirely by the weak side, that is by how far the small-field expansion can be pushed. The figure “about twenty doublings” quoted in the operator-inequality note §6 presupposes a weak side reaching $g^2 \sim 1/2$; with the crude constants of that note the region is fifty times wider.

5. Consequence for STATE

The T2 theorem stands, and its threshold, made explicit from the proof that supplies it, is $g_0^2 \sim 10^{101}$ for $SU(3)$, which covers nothing of physical interest. The strong-coupling boundary of the intermediate region has to be established by a direct expansion with explicit constants, expected to give $g_0^2 \sim 40\text{--}70$; that derivation is concrete, classical in method, and is the next step on the strong side. The width of the intermediate region is controlled by the weak side, and the relevant constant there is how far the small-field expansion reaches in $1/g^2$.